

Modeling the US Economy

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Based on *Linear Algebra and Its Applications* by D. Lay, et al.

Image Credit: <http://lasierrainformatica.blogspot.com/2013/06/el-harvard-mark-ii.html>

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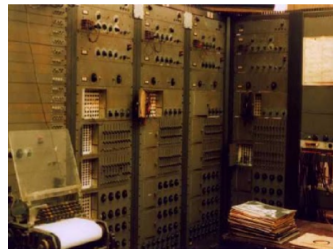
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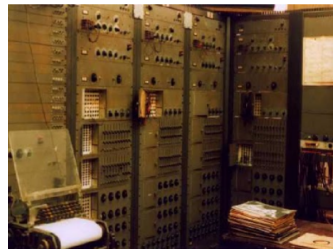


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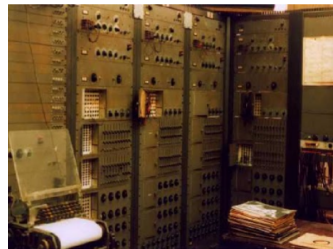


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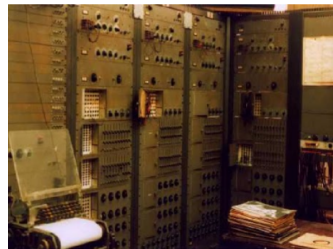


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- ▶ Mark II solved the system in 56 hours.



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		Amount Purchased by		
Sector	Total Output (in Billions)	Renewables	Electricity	Manufacturing
Renewable Energy	40	10 (25%)	25 (62.5%)	5 (12.5%)
Electricity	100	7 (7%)	18 (18%)	75 (75%)
Manufacturing	125	20 (16%)	50 (40%)	55 (44%)

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- ▶ How can we determine the equilibrium prices for each sector?

Setting Up a System of Linear Equations

Proportion of Output from:			
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- ▶ Down each column is the proportion of that sector which is purchased by each of the other three sectors.
- ▶ Across each row we see for a given sector, what proportion of their inputs came from each sector's output.
- ▶ For the renewable energy sector, we therefore have the following linear equation:

output from renewables = (input from renewables) + (input from electricity) + (input from manufacturing)

$$p_r = 0.25p_r + 0.07p_e + 0.16p_m$$

$$0.75p_r - 0.07p_e - 0.16p_m = 0$$

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Similarly deriving linear equations for the other sectors, we have the following system:

$$0.75p_r - 0.07p_e - 0.16p_m = 0$$

$$-0.625p_r + 0.82p_e - 0.4p_m = 0$$

$$-0.125p_r - 0.75p_e + 0.56p_m = 0$$

Finding the Equilibrium for the Economy

We have the following augmented matrix associated to the system:

$$\left[\begin{array}{ccc|c} 0.75 & -0.07 & -0.16 & 0 \\ -0.625 & 0.82 & -0.4 & 0 \\ -0.125 & -0.75 & 0.56 & 0 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & -0.28 & 0 \\ 0 & 1 & -0.70 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

Notice p_m is a free variable, and we have equilibrium solution:

$$\mathbf{p} = \begin{bmatrix} p_r \\ p_e \\ p_m \end{bmatrix} = \begin{bmatrix} 0.28p_m \\ 0.7p_m \\ p_m \end{bmatrix} = p_m \begin{bmatrix} 0.28 \\ 0.7 \\ 1 \end{bmatrix}$$

- ▶ If this economy has $p_m = 125$ billion dollars,
- ▶ Then if we want to ensure the economy is functioning at its equilibrium level (everything produced is used by other sectors):
 - ▶ Set $p_r = (0.28)(125) = 35$ billion dollars, and
 - ▶ Set $p_e = (0.7)(125) = 87.5$ billion dollars.

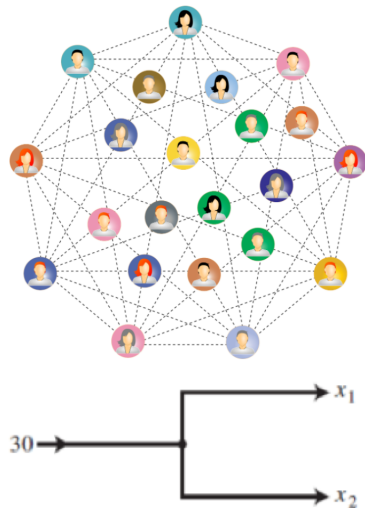
Network Flow

- ▶ A **network** consists of a set of points, called **nodes** with lines, called **branches** connecting some or all of the nodes.



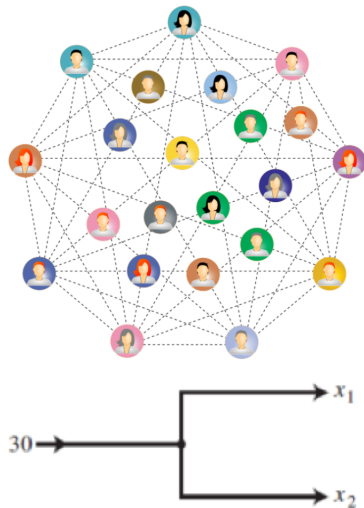
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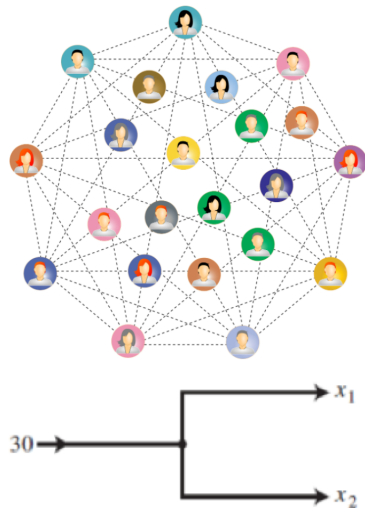
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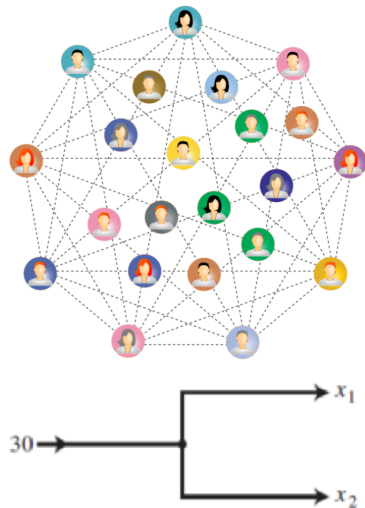
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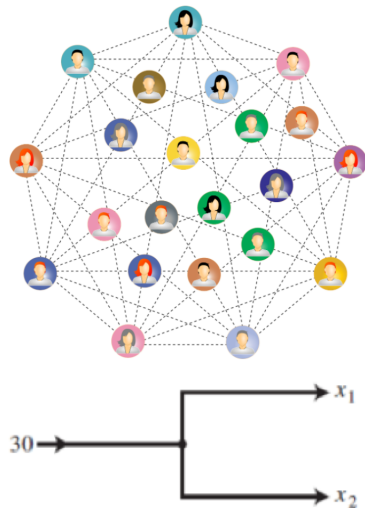
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- ▶ Network flows have applications to current flow through a circuit, flow of goods through supply chains, social networks, and **urban planning** to name a few.



Traffic Flow in Baltimore

The network in the figure shows the flow of traffic (in vehicles per hour) over several one way streets in downtown Baltimore during a typical early afternoon. Determine the general flow pattern for the network.

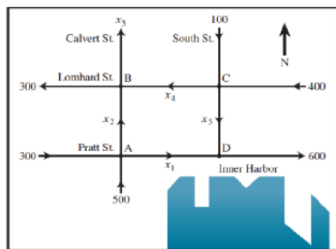


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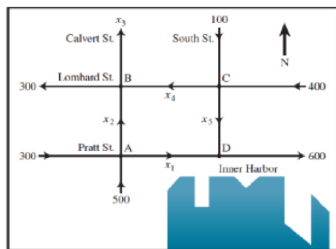


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Intersection	Flow in	Flow out
A	$300 + 500$	$= x_1 + x_2$
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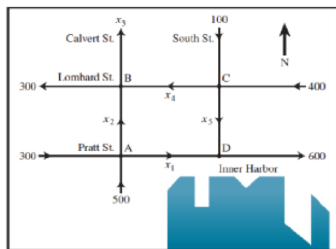


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$$\begin{array}{rcl}
 x_1 + x_2 & = & 800 \\
 x_2 - x_3 + x_4 & = & 300 \\
 x_4 + x_5 & = & 500 \\
 x_1 + x_5 & = & 600 \\
 x_3 & = & 400
 \end{array}$$

Solving the System

We need to solve the following nonhomogeneous linear system of equations:

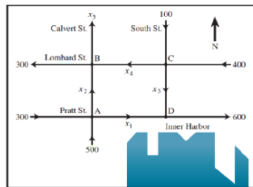


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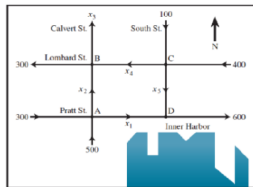


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We have an augmented matrix

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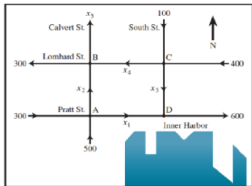


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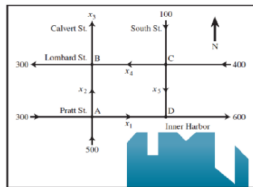


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